

YUNSUNG CHUNG

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SUMMARY

PhD candidate in Computer Science at Tulane University (anticipated graduation: December 2026) specializing in the evaluation, alignment, and deployment of machine learning systems in healthcare. My research focuses on building and validating models under sparse, noisy, and temporally misaligned clinical supervision, with an emphasis on safety, robustness, and outcome-oriented evaluation. My work spans wearable biosignals and medical imaging, including the use of generative and synthetic data methods (e.g., diffusion models) for alignment, data augmentation, and privacy-utility analysis. I have led end-to-end research projects from problem formulation through large-scale experimentation and reproducible publication at venues including USENIX Security, MICCAI, and ICASSP, with industry research experience at Samsung Research America and close collaboration with clinicians.

EDUCATION

Tulane University PhD in Computer Science	<i>Aug 2022 -</i>
The Australian National University M.S. in Computer Science	<i>Jul 2019 - Jul 2021</i>
Dongguk University B.S. in Electronics, Information Communication Engineering	<i>Mar 2012 - Feb 2018</i>

EXPERIENCE

Samsung Research America – Research Intern Mountain View, California	<i>May 2025 – Aug 2025</i> Supervisor: Dr. Sharanya Desai
Designed algorithms for predicting clinical biomarkers from large-scale wearable signal data (PPG) under sparse/noisy labels.	
Tulane University – Research Assistant, TRIAD Lab New Orleans, Louisiana	<i>Aug 2022 – Present</i> Supervisor: Dr. Jihun Hamm
- Lead projects on generative AI for synthetic medical data to address data scarcity and security, resulting in publications at USENIX Security and MICCAI. - Developed deep learning models for cardiac MRI, atrial fibrillation modeling, and wearable signal analysis, collaborating with clinicians across institutions.	
Airport Railroad Express – Hardware Engineer Incheon, South Korea	<i>Nov 2017 – May 2019</i>
Analyzed and tested hardware systems for automated fare collection and passenger service systems.	
Republic of Korea Army – Sergeant Incheon, South Korea	<i>Jan 2013 – Oct 2014</i>

PUBLICATIONS

Peer-Reviewed Conferences & Workshops

- [1] **Yunsung Chung**, Keum San Chung, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanhoo Lim, Viswam Nathan, Nassir Marrouche, and Sharanya Arcot Desai. *Weighted Temporal Decay Loss For Learning Wearable PPG Data With Sparse Clinical Labels*. **ICASSP**, 2026.

- [2] **Yunsung Chung**, Chanhho Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, and Jihun Hamm. *SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation*. **MICCAI**, 2025.
- [3] **Yunsung Chung**, Yunbei Zhang, Nassir Marrouche, and Jihun Hamm. *SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation*. **USENIX Security**, 2025.
- [4] **Yunsung Chung**, Janet Wang, and Jihun Hamm. *Bridging the Gap: Synthetic Data Augmentation through Inversion and Distribution Matching for Few-shot Learning*. **AI-ML Systems**, 2024.
- [5] Janet Wang, **Yunsung Chung**, Zhengming Ding, and Jihun Hamm. *From Majority to Minority: A Diffusion-based Augmentation for Underrepresented Groups in Skin Lesion Analysis*. **MICCAI Workshop**, 2024.
- [6] **Yunsung Chung**, Chanhho Lim, Chao Huang, Nassir Marrouche, and Jihun Hamm. *FBA-Net: Fore-ground and Background Aware Contrastive Learning for Semi-Supervised Atrium Segmentation*. **MICCAI Workshop**, 2023.

Manuscripts Under Review

- [7] **Yunsung Chung**, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, and Jihun Hamm. *CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis*. Under review at **ICML**, 2026.

Journal Articles

- [8] Han Feng, Chenguang Zhang, Ala Y. M. Assaf, Mario Mekhael, Chan Lim, Ghassan Bidaoui, Hadi Younes, Mayana Bsoul, Christian Massad, Yingshuo Liu, Yishi Jia, **Yunsung Chung**, et al. *Regional Volume and Regional Fibrosis Add Significant Information to the Prediction of Atrial Fibrillation Recurrence Using Machine Learning Model*. **Journal of the American College of Cardiology**, 2025.
- [9] Mayana Bsoul, Ala Y. M. Assaf, Ghassan Bidaoui, Hadi Younes, Christian Massad, Mohammad M. Atasi, Han Feng, Yingshuo Liu, Yishi Jia, **Yunsung Chung**, et al. *Outpatient Visits Decrease Stroke Risk in AF Patients*. **Journal of the American College of Cardiology**, 2025.
- [10] Chanhho Lim, **Yunsung Chung**, Jihun Hamm, Zhengming Ding, et al. *Generalizability and Clinical Implications of Electrocardiogram Denoising with Cardio-NAFNet*. **Journal of Biotechnology and Biomedicine**, 2024.

AWARDS & HONORS

FIRA RoboWorld Cup 2016 – 1st Place in Basketball	Dec 2016
International Robot Contest 2016 – 2nd Place in Marathon	Oct 2016
First-Class Academic Excellence Scholarship, Dongguk University	2012, 2015, 2016, 2017

EXTRA EXPERIENCE

Google Developers Machine Learning Bootcamp	Aug 2021 – Nov 2021
Humanoid Robotics Team – Gongdols	Mar 2015 – Dec 2016

SKILLS

Language - English, Korean, Python
Generative AI – diffusion models, synthetic data generation for data scarcity
Post-training of large pre-trained models - LoRA, preference-based optimization, alignment tuning
Trustworthy ML - privacy–utility evaluation, membership inference, differential privacy
Medical AI – cardiac MRI, ECG/PPG wearable signal modeling
Tools – PyTorch, MONAI, AWS, NumPy, Pandas, Scikit-learn, W&B